

Prompt Reflection Log

Assessment 5 - Telling the Story in Power BI | HR Analytics Dataset

This log reflects on how I used AI assistance during the development of the HR Analytics Power BI report. It records the questions I asked, how I evaluated and challenged the responses I received, and what I learned through the process of working through problems myself - with AI as a guide rather than a solution provider.

Prompt Log and Critical Reflections

Stage	What I asked	What the AI suggested	What I actually did/identified
Star Schema Design	"How do I convert a flat HR CSV into a star schema suitable for Power BI?"	Suggested splitting HR_Analytics.csv into Fact_Employee and 5 Dim tables using a Python script.	I ran the script in VS Code, verified the output files manually, and confirmed the row counts matched the original dataset. I identified that the script needed to be run from the correct folder path - something the initial instructions missed.
Data Quality Check	"The charts are all showing the same value of 314 - what is wrong and why is that? I think the dataset is wrong."	Suggested it was a DAX or relationship issue. Initial guidance was incorrect multiple times.	I identified that the root cause was actually the synthetic data I had been given - all categories were distributed perfectly equally, which is unrealistic. I flagged this myself and pushed back until the real issue was acknowledged and corrected.
Power Query Errors	"Getting an error: 'column AttritionRate was not found, why is it doing that?'"	Suggested updating the data source in Power Query.	I navigated Power Query myself, identified which tables had errors, and worked through the Applied Steps to find and remove the broken step. I also noticed that Dim_SalarySlab was missing from the query list and raised this independently.
DAX Measures	"How do I write an attrition rate measure as a percentage?"	Suggested <code>DIVIDE([Attrition Count],[Total Employees])</code> and formatting as a percentage.	I created each measure myself in Power BI, tested them against expected values, and identified when the Attrition Rate was showing 100%, which I correctly diagnosed as the card still pointing to the old flat file column, not the new measure.

Model View Relationships	"How do I create relationships between my Dim tables and Fact_Employee?"	Explained the dragging from PK in the Dim table to the FK in the Fact table with one-to-many cardinality.	I built all 5 relationships myself in Model View. I also noticed and questioned why HR_Analytics still appeared in the model with connecting lines, and independently removed those relationships before being told how to.
Salary Sort Order	"The salary slab chart is in the wrong order - how do I fix it?"	Suggested Sort by column in Column tools, pointing SalarySlab to SalaryOrder.	I applied this in Table View on Dim_SalarySlab myself. I also identified that the sort option wasn't appearing in the chart menu because the sort column hadn't been configured on the dimension table first - something I raised before the solution was fully explained.
GitHub Submission	"I already have a repository. How do I go about updating the current one, as I forgot some steps?"	Initial instructions assumed a new repository would be created.	I identified that the instructions were incomplete for my situation and asked for clarification. I then followed the corrected upload steps and managed the file organisation in my repository myself.

Overall Reflection

Throughout this project, I used AI as a sounding board and reference tool, but the analytical decisions, debugging, and critical thinking were my own. On several occasions, I identified errors in the guidance I received - including incorrect data, incomplete instructions, and Power BI steps that did not match what I was actually seeing on screen. Rather than accepting these at face value, I questioned them, tested alternatives, and pushed for corrections. This experience taught me that AI tools are most valuable when used critically - as a starting point for investigation rather than a definitive answer. The process of identifying what was wrong and knowing the right questions to ask is itself a core data analytics skill, and one I developed significantly through working on this project.